SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

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Tutorial-10_CO3
Subject Name: Discrete Mathematics
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Q.1 Which of these are propositions? What are the truth values of those that are propositions?
a) There is no pollution in Pune. b) What time is it? d) $4 + x = 5$. e) The moon is made of green cheese. f) $2^n \geq 100$

Q.2 Determine whether the following compound propositions are true or false.
a) $2 + 2 = 4$ if and only if $1 + 1 = 2$.
b) $1 + 1 = 2$ if and only if $2 + 3 = 4$.
c) $1 + 1 = 3$ if and only if monkeys can fly.
d) $0 > 1$ if and only if $2 > 1$.
e) If $1 + 1 = 2$, then $2 + 2 = 5$.
f) If $1 + 1 = 3$, then $2 + 2 = 4$.
g) If $1 + 1 = 3$, then $2 + 2 = 5$.
h) If monkeys can fly, then $1 + 1 = 3$.
i) If $1 + 1 = 3$, then unicorns exist.
j) If $2 + 2 = 4$, then $1 + 2 = 3$.
k) If $1 + 1 = 3$, then dogs can fly.
l) If $1 + 1 = 2$, then dogs can fly.

Q.3 Check the following propositions for Tautology, Contradiction, Contingency and Satisfiable.
a) $[p \wedge (p \rightarrow q)] \rightarrow q$
b) $(p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q)$
c) $\neg [(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)]$
d) $(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$

Q.4 Which of the following Propositions are logically equivalent.
a) $(p \rightarrow q) \wedge (p \rightarrow r)$ b) $(p \rightarrow r) \wedge (q \rightarrow r)$ c) $p \rightarrow (q \wedge r)$ d) $(p \vee q) \rightarrow r$

Q.5 Proof, without using a truth table that $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ is true when p, q, and r have the same truth value.

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